

A large, ancient tree with thick, gnarled trunks and a dense canopy of green leaves. Numerous thick, light-colored aerial roots hang down from the branches, some reaching the ground. The scene is set in a lush, green environment.

UNIT 2

PLANT PARTS AND THEIR FUNCTIONS

This tree has roots that hang above the ground.

Can the tree still get the water it needs to grow?

Let's Find Out:

- What are the functions of different plant parts?
- Why do some plant parts look different from others?

2.1

What Are the Functions of Different Plant Parts?

Roots, stem, leaves, flowers and fruit are the main parts of a plant.

Each part has its own function. They work together to get air, water and sunlight for the plant to stay alive and grow.

Leaves

The leaves trap sunlight and make food for the plant.

The leaves also have many tiny openings. These openings help the plant to take in and give out gases.

Stem

The stem supports the plant and holds the leaves up.

There are many tubes inside the stem. They carry water and minerals from the roots to other parts of the plant. They also carry food from the leaves to other parts of the plant.

Roots

The roots hold the plant firmly to the ground. The roots take in water and **minerals** for the plant. Plants need water and minerals to grow well.





Flower

The flower develops into a fruit.

Fruit

The fruit contains seeds. Seeds can grow into new plants.



CREATIVE SCIENCE

Make a 3D model of an apple tree!



Activity Book

► *Creative Science, pages 13-15*



WE CARE

Plants give us the oxygen we need to survive. Some plants also give us food to eat. It is important to care for plants.



Activity Book

► *Activity 1, pages 16-18*

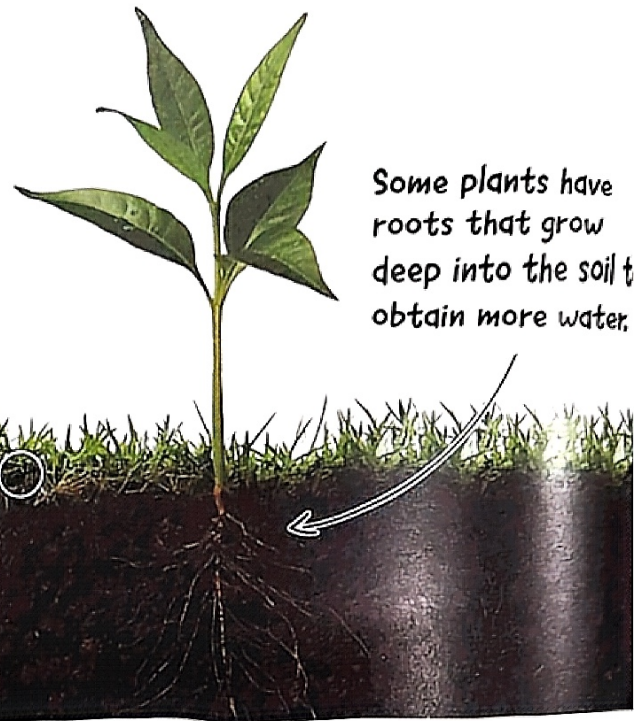
Why Do Some Plant Parts Look Different From Others?

Plants vary in size and shape. A plant part can look different in different plants. This is because the same plant part may have different functions in different plants.

Roots

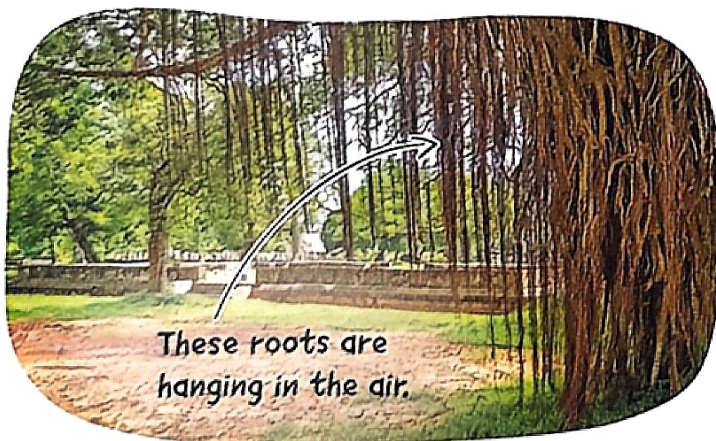
Roots are usually found underground.

Some plants have roots that spread over a wide area to obtain more water.



Some plants have roots that grow deep into the soil to obtain more water.

The roots of some plants are found above the ground.



These roots are hanging in the air.



These roots are sticking out of the mud.

These plants have roots above the ground to obtain enough air.

Stems

Stems are usually found above the ground.

Plants with strong stems, such as trees, can stand **upright**.



FLASHBACK

Recall from Textbook 1
Unit 5: Stems can be
weak or strong.



These trees have stems of different sizes and thicknesses. Their strong stems help them stand upright.

Plants with weak stems cannot stand upright. They use other plants, walls or fences for support, or just **creep** on the ground.



These plants have weak and long stems. Their long stems help them to climb or creep.

Leaves

The leaves of different plants have different sizes, shapes and **edges**.

The leaves of plants are usually thin and flat. This helps them absorb more sunlight to make food for the plant.

The leaves of tall trees get a lot of sunlight. These leaves are usually smaller.



The leaves of plants that grow under tall trees do not get as much sunlight. These leaves are usually larger.



This oak leaf is narrow with curvy edges.

Oak trees are tall. Their leaves can get more sunlight. The shape of the leaves helps the tree reduce loss of water due to heat from the Sun.



This dogwood leaf is wide with smooth edges.

Dogwood trees are shorter and further away from the Sun. They have wide leaves to help them absorb more sunlight and air.



RESEARCH

The Baobab tree is known as Africa's tree of life. Find out what makes it unique.

Flowers and fruit

Some plants have flowers. The flowers of different plants come in different shapes and colours.

These flowers are bright in colour.



These flowers are dull in colour.

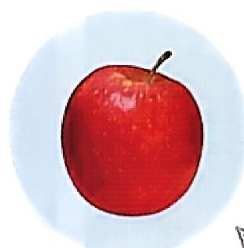


Flowers have parts that help plants **reproduce**. Different types of flowers attract different insects and animals that can help the plants reproduce.

Some flowers can develop into fruit. The fruit of different plants have different sizes, shapes and colours. Some types of fruit attract animals. The animals eat the fruit and the seeds fall to the ground and can grow into new plants.



This papaya is large, oval and yellow in colour.



This apple is small, round and red in colour.



QUICK CHECK

1. What are the main parts of a plant?
2. Why are the leaves of tall trees smaller than the leaves of shorter trees?
3. How do flowers and fruit help plants to reproduce?



FLASHBACK

Recall from Textbook 1 Unit 5: Some types of fruit have many seeds.



EXPLORE

What are some characteristics of fruit that can attract animals to eat them? Discuss with your classmates.



Activity Book

► Activity 2, pages 19-21

What We Have Learnt

PLANTS

have

**DIFFERENT
PARTS**

that can have different

Test Yourself

1. Which of the following is **not** correct?
 - (1) The leaves make food for the plant.
 - (2) The fruit of a plant develop into flowers.
 - (3) The roots take in water and minerals for the plant.
 - (4) The stem supports the plant and holds its leaves up.
2. Study these pictures.



Plant A



Plant B

Which of the following is correct?

- (1) Plant A has a weak stem.
- (2) Plant B has a strong stem.
- (3) Plants A and B have weak stems.
- (4) Plant A has a strong stem and plant B has a weak stem.

3. Study these plants.



Plant A



Plant B

- (a) Name **two** plant parts that can be found on plants A and B.
- (b) Name the plant part that provides support to these plants.

4. Ali has a plant in his garden. Part X of the plant helps it to reproduce. Part X develops into part Y. Part Y contains part Z.

- (a) Identify parts X, Y and Z.
- (b) Which part of the plant, X, Y or Z, can grow into a new plant?



Activity Book
Let's Review, pages 22-24



SCIENCE GLOSSARY

Creep

Grow along the ground

Edge

Side

Mineral

Substance that plants need to grow well

Reproduce

Produce more of its kind

Upright

Straight up

Science Today



Aeroponics is a way of growing plants without using soil. These plants are grown with their roots suspended in the air!

In aeroponics, the roots are directly sprayed with water containing minerals. Therefore, less water is needed for these plants to grow well compared to those grown in soil.

Factors like temperature can also be changed so that the plants can grow under ideal conditions. This allows them to be grown more quickly and successfully.

The plants can also be grown in layers that can be stacked upwards like shelves. This reduces the amount of space needed to grow them. More plants can be grown in a smaller space.



Basil, lettuce and mint can be grown using aeroponics.